PEH Data Platform – User Guidance

Table of contents

User Guide

> Tip
>
> **Download the full User Guide as PDF**
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This user guide describes how to use the PEH Data Platform in a correct and responsible way.

You can navigate to detailed guidance for:

Data Providers - Supplying Data Controllers

Data Users - Receiving Data Controllers

Protocol

Protocol On the management and use of the Personal Exposure and Health Data Platform

The Personal Exposure and Health Data Platform ("PEH Data Platform") has been developed at VITO in order to give access and to share single measurement Personal Exposure and Health data ("PEH data"), most commonly originating from population studies, in the highest possible resolution to researchers and other stakeholders.

It is the intention of the Supplying Data Controllers of the data included in this platform, to make the data on this data platform further available to stakeholders according to the terms and conditions of this protocol and the Regulation 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data ('GDPR', General Data Protection Regulation), and allow additional data to be added to it.

This Protocol describes the terms and conditions on the data exchange between Supplying and Receiving Data Controllers via PEH Data Platform, the role of the PEH Data Platform Manager, as well as the processing activities performed by the Receiving Data Controller.

The **Protocol On the management and use of the Personal Exposure and Health Data Platform** can be consulted at doi.org/10.5281/zenodo.19232226

Providing data to the PEH Data Platform and getting access to data from the PEH Data Platform is only possible upon signature of the Protocol On the management and use of the Personal Exposure and Health Data Platform

To initiate the signature process to sign the Protocol On the management and use of the Personal Exposure and Health Data Platform, send an e-mail to: PEHDataPlatform@vito.be

User account

Register for a user account

To apply for an account to the PEH Data Platform, go to https://peh.vito.be/ and follow the instructions.

Important note:

Providing data to the PEH Data Platform and getting access to data from the PEH Data Platform is only possible upon signature of the Protocol On the management and use of the Personal Exposure and Health Data Platform

Data Providers

Overview

Purpose

This section describes how to provide data to the **PEH Data Platform**.

It focuses on:

- required input structures
- data validation: supported checks and inspections
- known limitations

Conceptual workflow

Providing data to the PEH Data Platform typically follows these steps:

1. The data provider signs the Protocol
2. A codebook is prepared together with the data platform manager
3. The data platform build the project configuration in the PEH Data Platform
4. The data provider receives a tailored data template to provide the data
5. The data provider requests a user account to the PEH Data Platform
6. The data provider runs services in the PEH Data Platform on the filled out data template:

- data validation
- data enrichment
- summary statistics

7. The data provider inspects the outputs and upon approval, proceeds to data upload to the PEH Data Platform

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Limitations and responsibilities

> **!** Important
>
> The PEH Data Platform is a **supporting tool**.
> It does **not**: - guarantee regulatory or ethical compliance - detect conceptual errors in study design - replace formal data submission checks

Responsibility for data quality remains with the data provider.

Data requirements

Purpose of this page

This page describes the **data requirements** for using the **PEH Data Platform**.

It explains:

- which data structures are expected
- the role of templates and codebooks

This information is essential for correct and consistent use of the platform.

Requirements

The PEH Data Platform assumes that data providers:

- work with **predefined data templates** (dowload link will follow)
- follow **agreed codebooks** (click to download)
- take responsibility for data correctness and completeness

The platform supports inspection and harmonisation through data validation services, but does not enforce or modify data.

Codebooks

Codebooks describe the **meaning and coding** of variables, including:

- variable definitions
- allowed values or value domains
- units and categories where applicable

They form the reference against which input data are inspected during the data validation in the PEH Data Platform

> **Note**
>
> The PEH Data platform does not harmonise or recode values automatically.

Data Templates

What is a template?

A template defines:

Input data should:

Relationship between templates and codebooks

Templates and codebooks are complementary:

- **Templates** define *structure*
- **Codebooks** define *semantics*

Correct use of the PEH Data Platform requires alignment with both.

> **Important**
>
> Use of the PEH Data Platform does not shift responsibility for data quality.
> Data providers remain responsible for:
>
> - correctness of their data
> - alignment with project specifications
> - ethical and regulatory compliance

Data Users

Overview

Purpose

This section describes how to get access to data from the **PEH Data Platform**.

It focuses on:

- Submission of a **data request**
- Data extract codebook and format

Legal and ethical framework

Requests may involve **sensitive personal data**, including health data.
All processing must comply with applicable national and EU legal and ethical requirements.

All data requests are subject to the
Protocol on the management and use of the Personal Exposure and Health Data Platform.

Reference
Protocol on the management and use of the Personal Exposure and Health Data Platform. Zenodo. https://doi.org/10.5281/zenodo.19232226

Data requests

Data access request

Access to **individual-level data** stored in the **Personal Exposure and Health (PEH) Data Platform** is granted on request and subject to approval.

Who can request access?

Data access requests can be submitted by a **Receiving Data Controller (RDC)** for research projects in the public interest, in line with the objectives of the PEH Data Platform.

If multiple RDCs are involved in the same research project, and they require **different subsets of data**,
each RDC must submit a separate request, specifying: - their own research questions, and - the specific data requested.

Data access procedure

Step-by-step

1. The RDC submits a **Data Access Request Form**

- Via **Microsoft Forms**.
  - To preprare for a form submission, you may consult the corresponding word version here.
- Provides the required annexes as instructed in the form.
  - Selection of **studies**:

        - Selection of **variables**:

2. The request is reviewed by the **Data Request Access Committee (DRAC)**, including a data minimisation check.
   The original data providers (**Supplying Data Controllers – SDCs**) may object to the request.
3. If approved, access is granted:
   - to the requested subset of PEH data (studies and variables),
   - filtered to studies for which no SDC objection was raised, and
   - **only after signature of the Protocol** by the RDC (https://doi.org/10.5281/zenodo.19232226).
4. Approved data extracts are made available through the PEH Data Platform: https://peh.vito.be/

To avoid requesting data for research questions that have already been explored, it is advised to browse the Publications section in this user guide prior to submitting your request.

Information required in the Data Access Request Form

The Data Access Request Form collects the following information.

1. Project identification

- Main project name (acronym and full name)
- Research project title
- Date of submission

2. Scientific context

- Scientific background
- Objective and specific research questions

3. Methods

- Study design
- Target population
  (e.g. age groups, sex, inclusion/exclusion criteria)

4. Requested data

- Requested datasets and studies
- Requested variables

  Requested data **must be specified using the official annex files** generated via this application.

5. Data analysis plan

- Statistical methods and analytical models
- Sensitivity or pooled analyses (if applicable)
- Requirements for data comparability

6. Data minimisation

- Explicit declaration that only data strictly necessary to answer the research questions are requested

7. Administrative information

- Responsible person and institution
- Contact details
- Data analysts and (if applicable) external data processors
- Planned study start and end dates
- Publication intentions

Required annex files (generated via this app)

To complete a Data Access Request, **two annex files are required**.
Both must be **downloaded from this application**.

Annex – Studies

Download the **Annex – Studies** file(s) from Studies.

For each study you request: - set **`request_stud_1isYes` = 1**

All studies marked with `request_stud_1isYes = 1` are included in the request.

Annex – Variables

Download the **Annex – Variables** file(s) from the Codebooks

For each variable you request: - set **`request_var_1isYes` = 1**

All variables marked with `request_var_1isYes = 1` are included in the request.

Only annex files **generated and completed via the Studies and Codebooks sections of this application** are accepted.

Questions?

For questions regarding data availability or the data access procedure, please contact: **PEHDataPlatform@vito.be**

Publications

Approved Requests

Approved Data Access Requests

Request number	Title of request	Institution of submitter	Country of submitter	Resulting publication title	Date	Resulting publication doi
PEH_2023_01	Association between PFAS, phthalates/DINCH with Metabolic hormones, kisspeptin, and metabolic health outcomes	VITO	Belgium	NA	NA	NA
PEH_2023_02	Association between Arsenic, Metabolic hormones & kisspeptin, and metabolic health outcomes	VITO	Belgium	NA	NA	NA
PEH_2023_03	Association between exposure to acrylamide and the neurological biomarker BDNF	NIPH	Norway	NA	NA	NA
PEH_2023_04	Association between exposure to acrylamide and metabolism	NIPH	Norway	NA	NA	NA
PEH_2023_05	Association between UV filters, acrylamide and non-persistent pesticides and body mass index among adult population	INRAE	France	NA	NA	NA
PEH_2023_06	Association between pesticide exposure and asthma and allergies in children	EASP	Spain	NA	NA	NA
PEH_2023_07	Association between pesticide exposure and body mass index (BMI) in children	EASP	Spain	NA	NA	NA
PEH_2023_08	Association between arsenic exposure and reproductive hormones and sexual maturation	VITO	Belgium	NA	NA	NA
PEH_2023_09	Association between Cadmium with metabolism focusing on BMI in young adults	AU-PH	Denmark	NA	NA	NA
PEH_2023_10	Associations of PAHs with BMI	AUTH	Greece	NA	NA	NA

Request number	Title of request	Institution of submitter	Country of submitter	Resulting publication title	Date	Resulting publication doi
PEH_2023_11	Associations between mycotoxins exposure (DON) and body mass index in adults	SpF	France	NA	NA	NA
PEH_2023_13	Master's Thesis: Applying robust regression to assess exposure to bisphenols in the European population	INSA	Portugal	NA	NA	NA
PEH_2023_14	Association between UV filters, with metabolic and asthma/allergy alterations in teenagers	UGR	Spain	NA	NA	NA
PEH_2023_15	Association between Bisphenols and BMI in adults from HBM4EU Aligned studies	UGR	Spain	NA	NA	NA
PEH_2023_16	Epigenetic changes in relation to PFAS, phthalates, DINCH, and arsenic in the association between exposure and cardiovascular and endocrine effects	VITO	Belgium	NA	NA	NA
PEH_2023_17	Association of Arsenic, PFAS, Phthalates, and UV-filters on the neurological pathway: serum BDNF and thyroid hormones	VITO	Belgium	NA	NA	NA
PEH_2023_18	Association of UV-filters on kisspeptin and reproductive markers	VITO	Belgium	NA	NA	NA
PEH_2023_19	Derivation of European exposure values of internal exposure to environmental pollutants using human biomonitoring data – Method development	VITO	Belgium	NA	NA	NA
PEH_2024_01	Case study 7.3.2 Assessment of uncertainty in PBK models of PFAS exposure in humans and reverse dosimetry	MU	Czech Republic	NA	NA	NA
PEH_2024_02	Spatial determinants of PAHs exposure in European population	MU	Czech Republic	NA	NA	NA
PEH_2024_03	Spatial determinants of pesticide exposure	MU	Czech Republic	NA	NA	NA
PEH_2024_04	Feasibility study to evaluate occupational exposure in general surveys	TTL	Finland	NA	NA	NA
PEH_2025_01	PARC activity T4.1.4.1 on data management of HBM data	VITO	Belgium	NA	NA	NA

Publications

Overview of publications based on data available through the PEH platform

This page provides an overview of peer-reviewed publications that made use of data available through the PEH Data Platform.

HBM4EU AS studies

Publication title	Date	Publication doi
Characteristics paper	NA	NA
HBM4EU combines and harmonises human biomonitoring data across the EU, building on existing capacity - The HBM4EU survey	44431	https://doi.org/10.1016/j.ijheh.202
Harmonization of Human Biomonitoring Studies in Europe: Characteristics of the HBM4EU-Aligned Studies Participants	44713	https://doi.org/10.3390/ijerph1911
NA	NA	NA
Overview paper expoure levels	NA	NA
Harmonized human biomonitoring in European children, teenagers and adults: EU-wide exposure data of 11 chemical substance groups from the HBM4EU Aligned Studies (2014-2021)	44966	https://doi.org/10.1016/j.ijheh.202
NA	NA	NA
Exposure data analyses (determinants, trends, …)	NA	NA
Trends of Exposure to Acrylamide as Measured by Urinary Biomarkers Levels within the HBM4EU Biomonitoring Aligned Studies (2000-2021)	44775	https://doi.org/10.3390/toxics10080443
Glyphosate and AMPA in Human Urine of HBM4EU Aligned Studies: Part A Children	44785	https://doi.org/10.3390/toxics10080470
Time Trends of Acrylamide Exposure in Europe: Combined Analysis of Published Reports and Current HBM4EU Studies	44790	https://doi.org/10.3390/toxics10080481
Glyphosate and AMPA in Human Urine of HBM4EU-Aligned Studies: Part B Adults	44825	https://doi.org/10.3390/toxics10100552
Cadmium exposure in adults across Europe: Results from the HBM4EU Aligned Studies survey 2014-2020	44851	https://doi.org/10.1016/j.ijheh.202

Publication title	Date	Publication doi
PFAS levels and determinants of variability in exposure in European teenagers - Results from the HBM4EU aligned studies (2014-2021)	44865	https://doi.org/10.1016/j.ijheh.2022
Exposure to flame retardants in European children - Results from the HBM4EU aligned studies	44890	https://doi.org/10.1016/j.ijheh.2022
Human urinary arsenic species, associated exposure determinants and potential health risks assessed in the HBM4EU Aligned Studies	44947	https://doi.org/10.1016/j.ijheh.2023
Current exposure to phthalates and DINCH in European children and adolescents - Results from the HBM4EU Aligned Studies 2014 to 2021	44973	https://doi.org/10.1016/j.ijheh.2022
Time Patterns in Internal Human Exposure Data to Bisphenols, Phthalates, DINCH, Organophosphate Flame Retardants, Cadmium and Polyaromatic Hydrocarbons in Europe	45197	https://doi.org/10.3390/toxics11100819
Determinants of exposure to acrylamide in European children and adults based on urinary biomarkers: results from the "European Human Biomonitoring Initiative" HBM4EU participating studies	45262	https://doi.org/10.1038/s41598-023-48738-6
Exposure assessment of the European adult population to deoxynivalenol - Results from the HBM4EU Aligned Studies	45602	https://doi.org/10.1016/j.foodres.20
Polycyclic aromatic hydrocarbon (PAH) exposure among European adults: Evidence from the HBM4EU aligned studies	45732	https://doi.org/10.1016/j.envint.2025.109383
NA	NA	NA
Exposure-effect	NA	NA
Concurrent Assessment of Phthalates/HEXAMOLL® DINCH Exposure and Wechsler Intelligence Scale for Children Performance in Three European Cohorts of the HBM4EU Aligned Studies	44820	https://doi.org/10.3390/toxics10090538
Cross-sectional associations between exposure to per- and polyfluoroalkyl substances and body mass index among European teenagers in the HBM4EU aligned studies	44867	https://doi.org/10.1016/j.envpol.2022.120566
PFAS association with kisspeptin and sex hormones in teenagers of the HBM4EU aligned studies	45128	https://doi.org/10.1016/j.envpol.2023.122214
PFAS and Phthalate/DINCH Exposure in Association with Age at Menarche in Teenagers of the HBM4EU Aligned Studies	45156	https://doi.org/10.3390/toxics11080711
Association of exposure to perfluoroalkyl substances (PFAS) and phthalates with thyroid hormones in adolescents from HBM4EU aligned studies	45157	https://doi.org/10.1016/j.envres.2023.116897
Cognitive Performance and Exposure to Organophosphate Flame Retardants in Children: Evidence from a Cross-Sectional Analysis of Two European Mother-Child Cohorts	45223	https://doi.org/10.3390/toxics11110878
Urinary phthalate/DINCH metabolites associations with kisspeptin and reproductive hormones in teenagers: A cross-sectional study from the HBM4EU aligned studies	45397	https://doi.org/10.1016/j.scitotenv.2024.172426

Publication title	Date	Publication doi
Urinary concentrations of phthalate/DINCH metabolites and body mass index among European children and adolescents in the HBM4EU Aligned Studies: A cross-sectional multi-country study	45507	https://doi.org/10.1016/j.envint.2024.108931
Associations between Urinary Phthalate Metabolites with BDNF and Behavioral Function among European Children from Five HBM4EU Aligned Studies	45535	https://doi.org/10.3390/toxics12090642
Association of environmental pollutants with asthma and allergy, and the mediating role of oxidative stress and immune markers in adolescents	45619	https://doi.org/10.1016/j.envres.2024.120445
Adolescent exposure to benzophenone ultraviolet filters: cross-sectional associations with obesity, cardiometabolic biomarkers, and asthma/allergy in six European biomonitoring studies	45798	https://doi.org/10.1016/j.envres.2025.121912
NA	NA	NA
Risk assessment	NA	NA
Risk Assessment of Dietary Exposure to Organophosphorus Flame Retardants in Children by Using HBM-Data	44684	https://doi.org/10.3390/toxics10050234
Improving the Risk Assessment of Pesticides through the Integration of Human Biomonitoring and Food Monitoring Data: A Case Study for Chlorpyrifos	44721	https://doi.org/10.3390/toxics10060313
Cumulative risk assessment of five phthalates in European children and adolescents	30-Oct-2022	https://doi.org/10.1016/j.ijheh.2022
Approaches to mixture risk assessment of PFASs in the European population based on human hazard and biomonitoring data	44891	https://doi.org/10.1016/j.ijheh.2022

HBM4EU MOM-study

Publication title	Date	Publication doi
Characteristics paper	NA	NA
HBM4EU-MOM: Prenatal methylmercury-exposure control in five countries through suitable dietary advice for pregnancy – Study design and characteristics of participants	2023-07-01	https://doi.org/10.1016/j.ijheh.2023